

Test 1

October 4, 2023

Problem 1

Find each product or quotient:

1. $(-4)(-2)$

2. $(-3)(-9)$

3. $(4)(-6)$

4. $\frac{-49}{-7}$

5. $\frac{-16}{2}$

Problem 2

Evaluate each expression.

1. $\frac{1}{3} + (-\frac{4}{3})$

2. $\frac{3}{5} + \frac{5}{4}$

3. $\frac{11}{8} - \frac{1}{2}$

4. $\frac{9}{7} - (-\frac{5}{3})$

5. $(-2)(-\frac{5}{6})$

Problem 3

Evaluate each expression.

1. $\frac{-9-(2-5)}{-6}$

2. $[-1 - (-5)]|3 + 2|$

3. $(-7 - 5) \div [-2 - 2 - (-6)]$

4. $5(-5 + 6) \cdot 6^2$

5. $\frac{-13-2}{2-(-1)^3+(-6)-[-1-(-3)]}$

Problem 4

Evaluate: $p + (q - r)(6 - p)$ where $p = 6$, $q = 5$, and $r = 5$.

Problem 5

Solve: $\frac{r}{14} = \frac{5}{14}$.

Problem 6

Solve: $\frac{b}{3} + 7 = 10$.

Problem 7

Solve: $-7(x - 2) = -4 - 6(x - 1)$.

Problem 8

Solve: $-a - \frac{5}{4}\left(-\frac{8}{3}a + 1\right) = -\frac{19}{4}$.

Problem 9

Solve: $ax + bx = c$ for a .

Problem 10

Solve: $|5x + 3| = |2x - 1|$

Problem 11

Find the constant of variation and write the formula to express the relationship using that constant given that: t varies directly as the square of u , and $t = 6$ when $u = 3$.

Problem 12

A number plus itself, plus twice itself, plus 4 times itself, is equal to -104. What is the number?

Problem 13

A is now 34 years old, and B is 4 years old. In how many years will A be twice as old as B ?

Problem 14

A is 60 miles from B . An automobile at A starts for B at the rate of 20 miles an hour at the same time that an automobile at B starts for A at the rate of 25 miles an hour. How long will it be before the automobiles meet?

Problem 15

Sketch the graph of $y = -\frac{5}{4}x - 4$. CLEARLY label at least two points on the graph.

Problem 16

Find the slope of the line through the points $(-2, 10)$ and $(-2, -15)$.

Problem 17

Write the slope-intercept form of the equation $-3y = -5x + 9$.

Problem 18

Write the point-slope form of the equation of the line through the point $(-1, -5)$ with a slope of 9.

Problem 19

Write the slope-intercept form of the equation of the line through the point $(-2, 5)$, and perpendicular to the line $-x + y = -2$.

Extra Credit

An executive drove from home at an average speed of 40 mph to an airport where a helicopter was waiting. The executive boarded the helicopter and flew to the corporate offices at an average speed of 60 mph. The entire distance was 150 mi. The entire trip took 3 hrs. Find the distance from the airport to the corporate offices.